

Lannik Psi antenna

Technical description for RFQ · 17 September 2026
Saferadar Research · For supplier discussion and quotation

1. Product concept and scope

Lannik Psi is a 76–77 GHz FMCW radar with eight transmit (TX) ports and eight independent receive (RX) channels. We seek antenna development and prototype characterization for two variants, using a common TX concept and different RX layouts. This document describes the intended operation, the starting design and the information needed to develop it together.

The antenna has two complementary roles: **provide sensitivity for long-range detection, and provide angular information to resolve ambiguous directions.** Large RX subarrays improve receive gain, but their channel spacing means that some different directions produce similar or identical RX responses. The radar therefore periodically changes how the TX aperture is driven. The resulting TX responses give the processor additional information about target direction.

Three operating configurations on the same antenna



Front view: radar left is page right. Four groups × two ports = eight TX ports.
Colors show waveform assignment; internal feed boundaries are to be designed.

Figure 1: TX waveform assignments. UL/UR/LL/LR identify physical upper/lower and left/right groups. Each group is fed by two MMIC ports.

In **coherent operation**, all eight ports transmit the same waveform with nominal phase settings, producing the main sensitivity beam. In **left/right MIMO**, each half sends a separable waveform; the receiver measures both responses at every RX channel. **Up/down MIMO** uses the corresponding vertical split. The two MIMO configurations are interlaced with coherent operation; the exact timing and waveform implementation are radar responsibilities.

The antenna must support these combinations through its independently fed aperture groups. Within a waveform group, fields add coherently. The relative amplitudes **and phases** of the separated group responses vary with direction and are used by signal processing. Consequently, the individual group patterns matter alongside the coherent sum beam. This concept does not require an antenna switching network.

What we are asking the supplier to quote

Please propose antenna/feed design, simulation, prototype manufacture and characterization for both RX variants. Identify shared and variant-specific work, assumptions, exclusions, costs and lead times. We expect an early technical review of feasibility and predicted patterns, followed by agreement on detailed targets and verification. Sections 2–4 describe the starting point; section 5 distinguishes quotation inputs, early design evidence and final data. Appendices contain coordinate, metric and data-format details.

2. Two RX variants and the geometry baseline

Each RX channel receives through its own subarray. The large variant uses rectangular subarrays in four columns of two channels, with alternating column heights. The small variant uses square subarrays in two columns of four channels. Both retain all eight independent RX channels for digital processing.

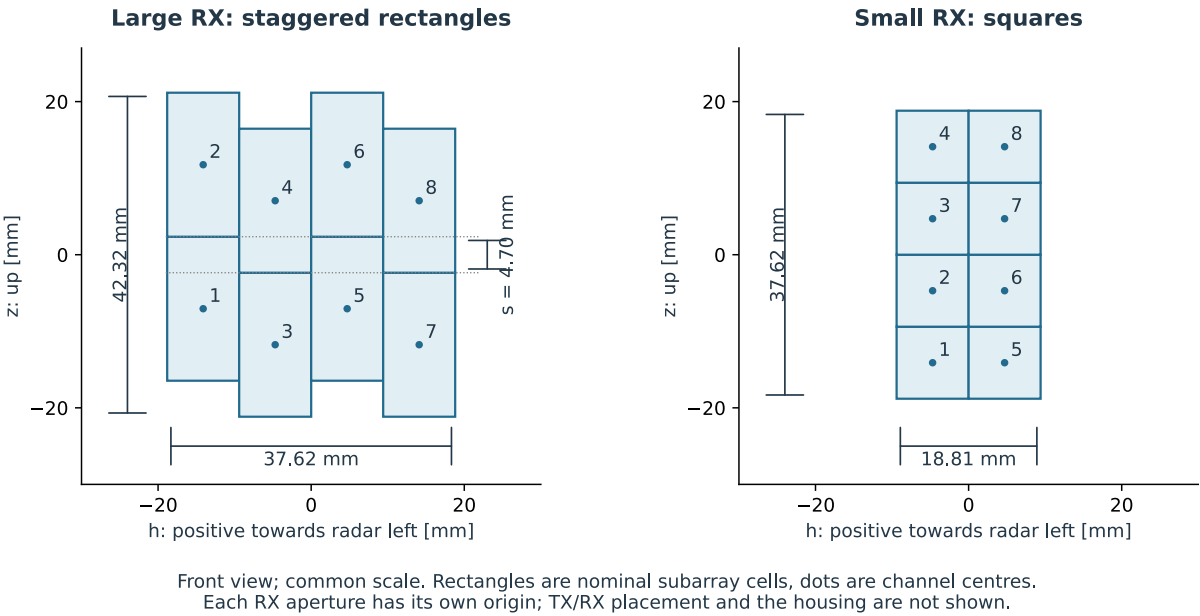


Figure 2: Selected RX layouts, with logical channel numbers matching Appendix A. The large layout has an alternating column stagger of $H/4$.

Nominal geometry	Large variant	Small variant
Channel arrangement	4 columns $\times$ 2 channels, staggered	2 columns $\times$ 4 channels
Subarray cell, horizontal $\times$ vertical	9.405 $\times$ 18.811 mm	9.405 $\times$ 9.405 mm
Difference in height between adjacent columns	4.703 mm	None
Overall RX cell envelope	37.62 $\times$ 42.32 mm	18.81 $\times$ 37.62 mm
Reference radiators per channel	4 $\times$ 8, uniformly excited	4 $\times$ 4, uniformly excited

The large variant retains more RX aperture and gain. Its stagger gives ordinary coherent measurements useful information about some vertical ambiguities. MIMO remains part of both variants, including for horizontal ambiguity and for the small variant’s vertical ambiguity.

How to use these dimensions

These dimensions are the **analysis baseline**, derived from Saferadar’s reference radiator layout at approximately 2.351 mm pitch. The cell dimensions and channel spacings were constructed from that layout; their quoted precision does not establish manufacturing tolerances. They are neither radiator-edge measurements nor final mechanical outlines.

We want to preserve the behavior demonstrated by that baseline. **Reasoned adjustments to dimensions, pitch, radiator construction or feed layout are welcome for joint evaluation.** Please show proposed changes explicitly, with their rationale and predicted complex patterns. We will assess the effect on gain, coverage, resolution and ambiguity before adopting the revised geometry. Even a small dimensional change should be visible in the design data; it need not be ruled out in advance.

The diagrams show each RX aperture in its own coordinate system. TX/RX aperture and subarray dimensions, spacing and feed geometry remain in scope. Mounting and other mechanical interfaces are defined separately. Approximate overall dimensions are in section 4; electrical coordinates are in Appendix A.

3. TX concept and reference excitation

The starting TX architecture has four independently driven aperture groups, each fed by two equal-power MMIC ports. The supplier is asked to design the partition and feed network so that all eight ports can operate at nominal full power. The baseline uses separately fed aperture regions whose fields combine; it does not call for an RF combiner joining two MMIC outputs. Geometrically equal port areas are not necessary.

The reference below uses a tapered amplitude distribution and a smooth quadratic phase distribution. The phase shaping broadens the coherent beam and produces different left/right and up/down group responses. Its nominal TX cell envelope is approximately **37.62 × 37.62 mm**; Appendix A gives the reference pitch and distinguishes cell dimensions from radiator-centre spans.

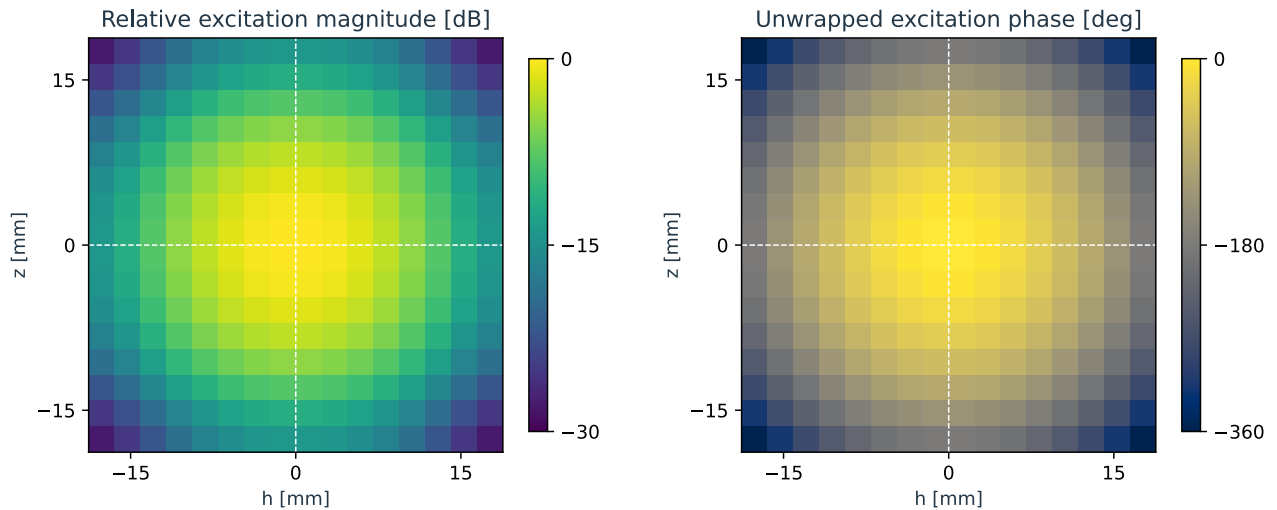


Figure 3: Reference TX excitation at 77 GHz. Phase is unwrapped and referenced to the central radiators; dashed lines show the four conceptual groups.

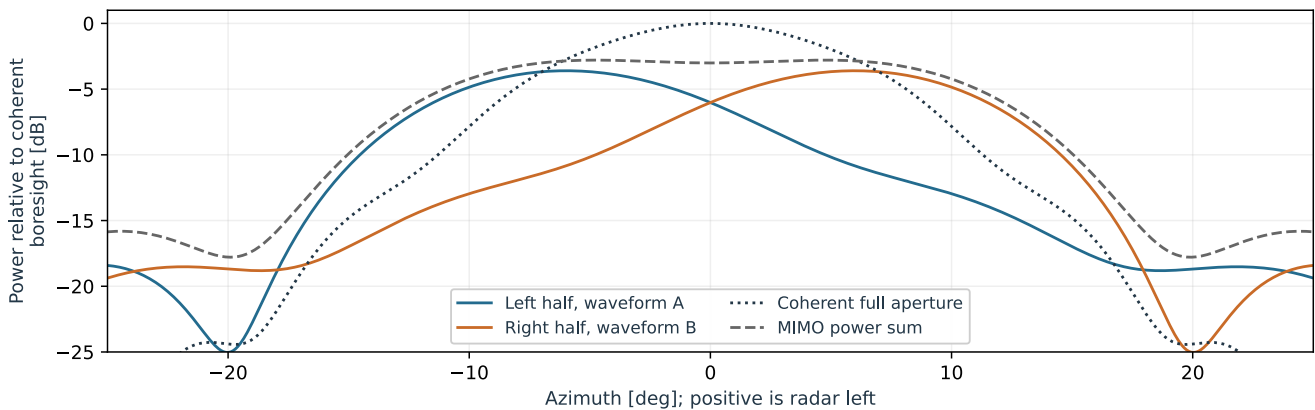


Figure 4: Illustrative horizontal cut at 77 GHz, with the same total TX power in both modes. Each half curve includes its half share of that power.

In this ideal model, the coherent beam is about 12.6° wide at half power; the two half beams peak roughly 6° to either side of boresight. The modeled boresight gain is approximately 23.5 dBi, using an inferred radiator gain. These figures do not include a verified feed-loss or installed-antenna model.

The 16 × 16 radiator count, exact pitch, taper law and squint are reference implementation details. Alternative implementations can be proposed and evaluated using the complete patterns. Matching only the sum-beam gain and width does not establish the required MIMO behavior. The up/down split serves the same purpose in elevation and must also be characterized.

Tapering should redistribute available power without intentional dissipative attenuation or systematic TX backoff. Actual feed losses and the directivity cost of beam broadening must be reported. The reference amplitude taper's 35 dB Taylor parameter is not a guaranteed sidelobe level of the phase-shaped beam. Appendix A describes the accompanying reference excitation file.

4. Baseline, targets and design freedom

Confirmed operating band: 76–77 GHz, nominal centre frequency 76.5 GHz. The front end is the Infineon CTRX8188F with eight TX and eight RX channels. TX uses equal nominal full power per port with independent phase/waveform control. Use a common linear polarization for TX and RX: **either vertical or horizontal, with the choice to be confirmed jointly.**

The architecture and selected RX arrangements form the quotation baseline. The following numbers are **proposed discussion targets**, for supplier feedback on achievable performance and tradeoffs. They are not frozen production acceptance limits. Reference figures and detailed coordinates describe the starting model, with geometry changes handled as described in section 2.

Directional performance

All beamwidths below are **full, one-way 3 dB widths**, azimuth × elevation.

Quantity	TX, common concept	Large RX, per channel	Small RX, per channel
Boresight directivity	≥23 dBi	≥21 dBi	≥18 dBi
Approximate beamwidth	13° × 13°	20° × 10°	20° × 20°
Realized gain	Report for coherent and both MIMO modes	Report each channel	Report each channel

Gain and beamwidth must be assessed together. The main TX beam supports long-range sensitivity; responses around ±12° in the principal planes also matter for nearer targets and ambiguity resolution. Neither the beamwidth nor the characterization grid defines a hard product field of view. Report main-beam shape, sidelobes and nulls in 2-D so that useful two-way coverage can be assessed.

RF targets and approximate overall dimensions

Item	Target / indicative dimension
Large antenna overall dimensions (indicative)	Approximately 100 × 60 × 10 mm
Small antenna overall dimensions (indicative)	Approximately 80 × 70 × 10 mm
Port match across the band	Return loss ≥20 dB: $20 \log_{10}(\text{abs}(S_{ii})) \leq -20$ dB
TX–RX isolation	≥35 dB between every TX/RX port pair
TX–TX and RX–RX isolation	≥15 dB between distinct ports of each type
Radiation efficiency	≥80%, including feed dissipation, referred to accepted input power
RX boresight channel gain balance	Each channel within ±1 dB of the channel mean, before calibration

Please identify targets that materially constrain feasibility or cost. Report **realized gain at agreed input reference planes**, including feed and mismatch losses, alongside directivity and efficiency. RX digital combining gain must be identified separately from single-channel antenna gain. Pairwise isolation also needs review against active match and total RX leakage with multiple TX ports driven; supply the relevant coupling data.

Exact mechanical interfaces are covered by separately defined documentation. For the electrical assessment, state the modeled/measured installation, temperature, power/duty cycle and sample quantities assumed in the quotation. Final pattern masks, electrical tolerances and acceptance methods will be agreed after the preliminary design evidence is available.

5. Information and collaboration through the design

We propose the following technical review sequence. Please distinguish the scope and cost of each phase in the quotation, including measurement work.

With the quotation: establish the scope and approach

Describe the proposed antenna/feed implementation, compatibility with the three TX configurations and two RX layouts, and any deviations from the baseline. Give expected gain/loss and packaging feasibility, the difficult tradeoffs, and the planned simulation, measurement and calibration methods. State which assessments are estimates and which already have supporting data.

Detailed new electromagnetic simulations are not assumed to be part of preparing the quotation. Please identify the development work needed to produce the early evidence below, the proposed review points, and any assumptions that could change cost or schedule.

Early design review: establish that the concept will work

Before committing to fabrication, we need enough information to evaluate the implemented concept and agree the detailed targets:

- Proposed aperture/subarray dimensions and physical outlines, feed/port mapping, relative TX drive phases, and any changes from the baseline.
- **Simulated complex patterns of each TX port and RX channel**, with common phase references and defined power normalization, so that we can reconstruct the operational modes. Early group-level patterns may support a first feasibility review while the internal partition is being developed; the actual port patterns are needed when that partition is defined.
- Predicted absolute gain, feed/mismatch losses and relevant S-parameters over the band, plus principal pattern behavior and known model omissions.
- Initial sensitivity to dominant manufacturing and calibration errors, and a realistic plan for prototype measurements and production calibration.

An agreed small sample data file should establish conventions before the full export. Sampling and measurement scope can be refined together (Appendix C). Saferadar will assess two-way gain and competing angular responses for both RX variants and both MIMO splits, then return the limiting cases for discussion. Appendix B gives compact diagnostics the supplier may also use during design.

Final prototype/design delivery: supply the realized antenna description

Provide the **as-built geometry and port mapping**, nominal drive settings, simulated and measured complex port/channel patterns, absolute gain/loss data and S-parameters in the agreed formats. Include measurement conditions, reference planes, calibration/de-embedding, uncertainty, repeatability and any simulation-to-measurement differences. Identify the tested hardware, radome/housing configuration, temperatures, unit/sample coverage and revisions.

Explain manufacturing variation, residual drift after calibration and what must be measured or calibrated per production unit. Full angular measurement of every production unit is not assumed; propose a practical approach. If an agreed dataset cannot be measured, identify the gap and propose how the remaining uncertainty will be bounded and reviewed.

These data will update our range/coverage models and the angle-dependent responses used by signal processing. Predictable channel differences can often be modeled; unknown changes between the modeled and actual antenna are more problematic. We need both the nominal responses and a defensible description of their variation. Final antenna verification and system integration will use the agreed criteria; waveform processing and tracking remain Saferadar's scope.

Appendix A. Coordinate and excitation reference

A.1 Axes and nominal RX centres

Use x forward along boresight, y left when looking along boresight and z up. Azimuth is positive left; elevation is positive up. Direction cosines are $u = \sin(\text{azimuth}) \cos(\text{elevation})$ and $v = \sin(\text{elevation})$.

The table uses $h = y$ and z , with a separate origin at each RX aperture's centre. In a **front view**, looking back at the antenna face, positive h appears to the viewer's right. Document this transformation and channel-to-MMIC mapping consistently with the separately defined interfaces. Numbers are logical channel labels.

The reference pitch is approximately $p = 2.351$ mm. Nominal cell dimensions and spacings are $W = 4p \approx 9.405$ mm and $H = 8p \approx 18.811$ mm. For the large variant, columns in increasing h have offsets $(+s/2, -s/2, +s/2, -s/2)$ with $s = H/4 = 2p \approx 4.703$ mm. The small variant has spacing w in both axes.

Channel	Large h [mm]	Large z [mm]	Small h [mm]	Small z [mm]
1	-14.11	-7.05	-4.70	-14.11
2	-14.11	+11.76	-4.70	-4.70
3	-4.70	-11.76	-4.70	+4.70
4	-4.70	+7.05	-4.70	+14.11
5	+4.70	-7.05	+4.70	-14.11
6	+4.70	+11.76	+4.70	-4.70
7	+14.11	-11.76	+4.70	+4.70
8	+14.11	+7.05	+4.70	+14.11

These coordinates preserve the analyzed baseline. They do not require a frequency-independent electrical phase centre or prescribe dimensional acceptance tolerances. Propose tolerances based on their electrical effects; previous indicative position allowances such as ± 0.2 mm have not been validated as acceptance limits.

A.2 TX excitation file

The accompanying reference `antenna_arr_77_TX_rev_A.mat` contains MATLAB variable `antenna_arr`, one row per radiator. Columns 2 and 3 are horizontal and vertical positions in metres; column 5 is relative complex excitation; column 6 contains reference group labels. Both excitation magnitude and phase matter. Confirm position, viewing and phasor conventions when importing it.

The file represents a 16×16 radiator layout, approximately 37.62×37.62 mm on a cell-envelope basis (outer radiator centres span approximately 35.27 mm). It supplies the reference field distribution, **not a finalized eight-port feed partition**. Its group labels must not be interpreted as binding MMIC port boundaries. The supplier should propose that partition, nominal relative port phases and any resulting changes to the fields.

The quoted dimensions originate in this reference representation. There is no independent requirement for their numerical precision or for the same radiator count in an alternative realization. Document proposed adjustments and provide updated patterns for the joint assessment described in section 2.

Appendix B. Angular discrimination diagnostics

B.1 Direction-pair correlation

For one MIMO configuration, let $\mathbf{t}(d) = [T_1(d), T_2(d)]$ contain the two complex waveform-group fields at direction $d = (u, v)$. Sum the driven port fields with their nominal amplitudes and phases. Preserve relative group power; do not independently normalize group peaks. Define

$$\rho_{TX}(d1, d2) = \frac{\text{abs}(\mathbf{t}(d1)^H \mathbf{t}(d2))^2}{(|\mathbf{t}(d1)|^2 |\mathbf{t}(d2)|^2)}$$

$$\rho_{TX_dB} = 10 \log_{10}(\rho_{TX})$$

H denotes conjugate transpose. At 0 dB the two responses are proportional and cannot distinguish the directions with unknown target amplitude/phase. More-negative values indicate better ideal separation. This is spatial response correlation, distinct from waveform correlation and port isolation. Correlation is undefined for a zero-power response; report low-signal cases explicitly rather than treating them as successful discrimination.

B.2 Proposed checks for the baseline geometry

At each frequency, define $\lambda = c/f$, $a = \lambda/(2W)$ and $b = \lambda/(2H)$ using Appendix A. At 77 GHz, $(a, b) \approx (0.20698, 0.10349)$; at 76.5 GHz, approximately $(0.20833, 0.10417)$. These are direction cosines.

Direction pair (u,v)	MIMO split	Proposed ρ_{TX} limit	Ideal 77 GHz reference
(+a,0) / (-a,0), both RX variants	Left/right	≤ -2.5 dB	-3.1 dB
(0,+a) / (0,-a), small RX	Up/down	≤ -2.5 dB	-3.1 dB
(0,+b) / (0,-b), large RX	Up/down	≤ -10 dB	-29.3 dB

These are **discussion targets across 76-77 GHz**, not a complete acceptance test. Reference results use ideal 77 GHz excitation weights and do not verify physical performance across the band. Report directional group-power sums with each correlation. Update the evaluation directions when geometry changes; for the staggered layout, the old vertical alias is already a near-alias.

B.3 System assessment and robustness

Saferadar will evaluate the full response $A[g, r](d) = T_g(d) R_r(d)$, using actual RX patterns and receiver-noise weighting. At exact ideal RX aliases, TX correlation describes the remaining discrimination. Elsewhere, including staggered RX and displaced competing lobes, the full response is needed.

Correlation is considered alongside received signal strength. For a fixed single-target signal, a useful diagnostic in whitened noise is $D(d1 \rightarrow d2) = \text{SNR}(d1) * (1 - \rho_{full}(d1, d2))$, with linear SNR. It measures signal energy not explained by the wrong direction after fitting complex amplitude. It is not a correct-resolution probability. Evaluate both directions of a pair because their gains can differ. At -10 dB, $1 - \rho = 0.90$; pursuing the reference's very deep minimum is less important than preserving useful signal and separation across the relevant region.

The review searches competing lobes in 2-D, excluding local angle-estimation offsets, and states any gain/target-strength screening. It includes actual responses processed with the intended calibrated model, not only correlations computed with perfect knowledge of each antenna. A $\pm 15^\circ$ residual phase budget between alternating RX column groups is an initial system target to evaluate, not a raw antenna tolerance. Coverage, wrong-cell probability, latency and remaining calibration budgets are system-level decisions still to be finalized.

Appendix C. Pattern and characterization data

The following describes the intended early simulation and final delivery data. Agree the export schema using a small sample file and identify achievable measurement scope, limitations and costs in the proposal.

C.1 Reference conventions and contents

- Provide complex embedded fields for each of eight TX ports and eight RX channels, including co- and cross-polar components. State the other-port terminations and reference impedance. Use a common spatial origin and consistent amplitude/phase references across ports, with defined units, input-power normalization, phasor convention and polarization basis.
- Preserve spatial phase and relative magnitude. Do not independently recenter or peak-normalize port patterns. State reference planes, fixture removal, calibration/de-embedding and any correction from measurement to operational loads. RX data may use reciprocal transmit measurements with the conversion convention documented.
- Supply machine-readable data as well as plots, indexed by hardware revision, port/channel, frequency, direction, polarization and test condition. Identify simulated and measured datasets separately. Document any omitted feed, coupling, radome or housing effects.
- From nominal drive settings, provide the coherent sum beam, both MIMO group fields and their power sums. Compare modes at the same total incident TX power with their physical allocation retained. Include absolute directivity, realized gain, efficiency, beamwidth, sidelobes and nulls, and relevant S-parameters including TX-RX coupling.

C.2 Initial sampling proposal

For detailed patterns, start with $|u|, |v| \leq 0.6$ at spacing no coarser than 0.005 in each coordinate, at 76, 76.5 and 77 GHz. Equivalent angular grids or a demonstrated accurate interpolation method can be agreed. These are characterization coordinates, not a promised product field of view.

Include broader forward-hemisphere sidelobe/grating-lobe coverage at a suitable coarser spacing and refine significant lobes. Supply swept S-parameters across the band and additional pattern frequencies wherever dispersion or resonances require them. Three sampled frequencies alone cannot establish performance between them. An early reduced dataset is useful when its limits are stated; agree final coverage and uncertainty before the measurement campaign.

C.3 Variation, calibration and final interpretation

Distinguish constant complex port offsets, calibratable frequency dependence, and residual angle-dependent or time/temperature-dependent changes. Known repeatable differences can be represented in our models and processing. Calibration cannot recover lost signal power or eliminate unmodeled pattern changes; gain and model accuracy both matter.

Provide sensitivity to dominant feed, radiator, assembly and temperature variations, including correlated group errors. State measurement dynamic range and uncertainty; relative amplitude/phase tolerances at deep pattern nulls are not meaningful. Port phase errors within a coherent TX group can change the radiated pattern itself.

For final delivery, relate measurements to the as-built geometry and the proposed production calibration. Identify the sample count and tested conditions, remaining unmeasured variation, and which response parameters require per-unit calibration. Clearly separate bare-antenna predictions from measured or modeled installed behavior. These distinctions are needed to update system models and implement the correct angle-dependent response in signal processing.